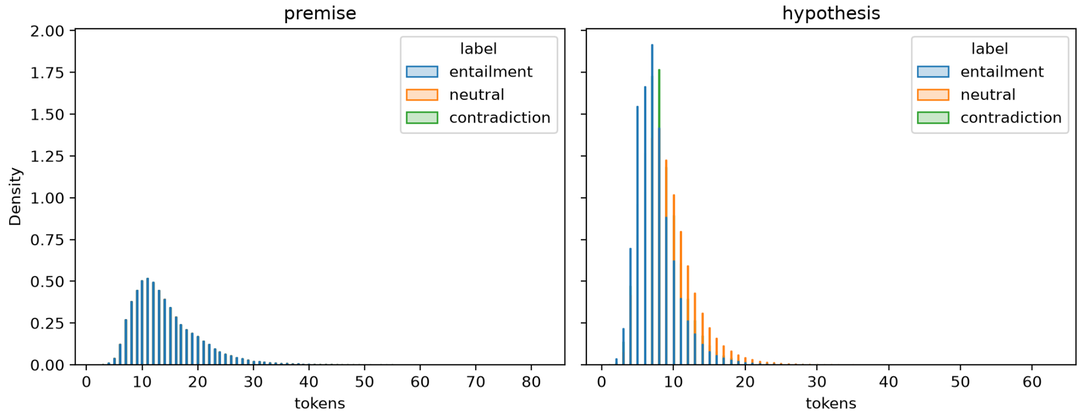
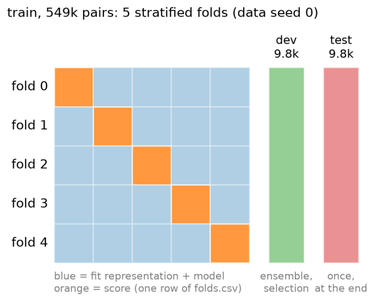
NLP1 Project 1 — NLI on SNLI

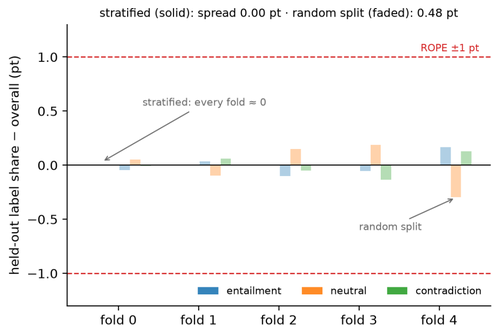
Emil Halliday · Neel Shah
6 Nov 2026
github.com/neelshahx/nlp1-2026

Q1 · Task & data

- (premise, hypothesis) → entailment / neutral / contradiction
- SNLI: 549k train / 9.8k dev / 9.8k test, balanced (~33% each)
- Artifacts (Gururangan et al. 2018): negation → contradiction, longer neutral hypotheses
- Token overlap (Jaccard): ent 0.31 · neu 0.26 · con 0.22
- PMI(token, label) = log p(token | label) / p(token)
- Setup: stratified 5-fold CV on train (seed 0); dev for selection; test once
- Stratified: every fold keeps the 33/33/33 label mix, so fold scores differ by model, not by label mix (random split drifts ±0.5 pt, half the ROPE); same folds for all → paired comparisons



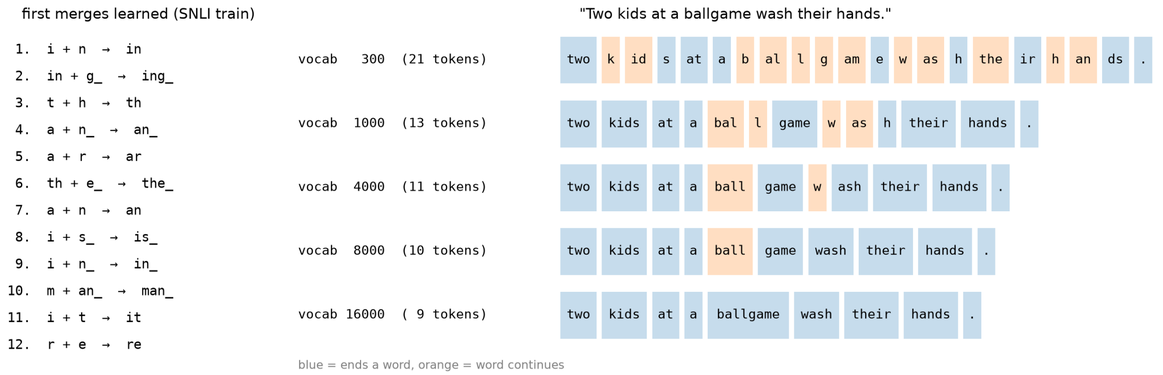




Every experiment scores the same 5 stratified folds (paired). Right: each held-out fold's label share minus the overall share. Stratified folds sit at 0; a random split of the same data drifts by up to 0.5 pt, half the ROPE.

Q2 · Tokenizer

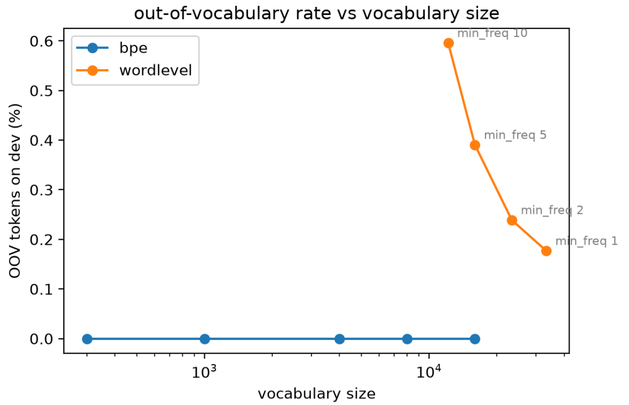
- BPE (HF tokenizers), min_freq 2, trained on SNLI train (premise + hypothesis) only
- 549k pairs is enough data: no external corpus needed
- Vocab 4k / 8k / 16k → CV acc 64.0 / 64.2 / 63.9: flat
- Word-level baseline 63.9 vs BPE 64.3 → inside ROPE; BPE wins on coverage, not accuracy
- Subwords → 0% out-of-vocabulary on dev (word-level 0.2–0.6%)
- Tokenizer is fit on all of train (incl. each held-out fold), but uses no labels



Left: BPE merges most frequent adjacent pair first. Right: one dev hypothesis at growing vocabulary sizes; rare word last to merge.

Premise: “Two young children in blue jerseys, one with the number 9 and one with the number 2 are standing on wooden steps in a bathroom and washing their hands in a sink.”

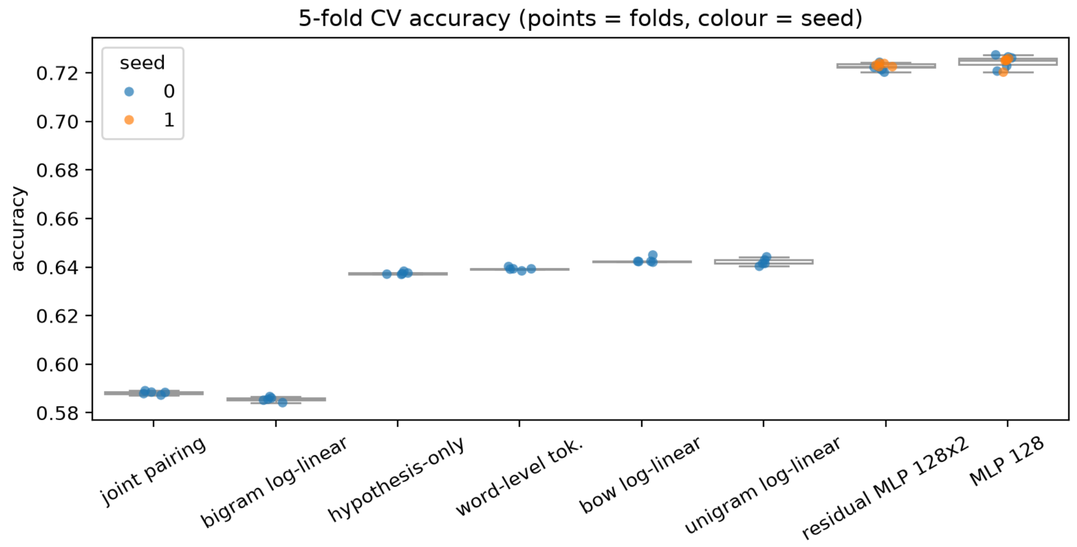
Hypothesis-only: entailment at 0.35 / 0.33 / 0.31, a coin flip; best log-linear: entailment; MLP: neutral (0.54) = truth.



Share of dev tokens the tokenizer maps to [UNK]. BPE falls back to characters so it never fails; word-level cannot.

Q3 · Representation

- Both sides at once: concat [BoW(P) ; BoW(H)], so the model knows which side a token is on
- Joint (one bag over "P [SEP] H" = BoW(P) + BoW(H)) 58.8 vs concat 64.3: –5.5 pt, side information lost
- Hypothesis-only 63.7 vs concat 64.3, P(rope) = 0.997
- Indicators vs counts: 64.28 vs 64.24 → inside ROPE (short sentences)
- Linear on concat = f(P) + g(H): no P × H interaction
- n-grams: unigrams 64.2 ≈ BoW; bigrams 58.6 per fold (–5.7, overfit sparse bigrams) yet 67.8 as a 5-fold ensemble on dev
- Next: TF-IDF, cross features |u – v|, u ⊙ v, MLP on bigrams

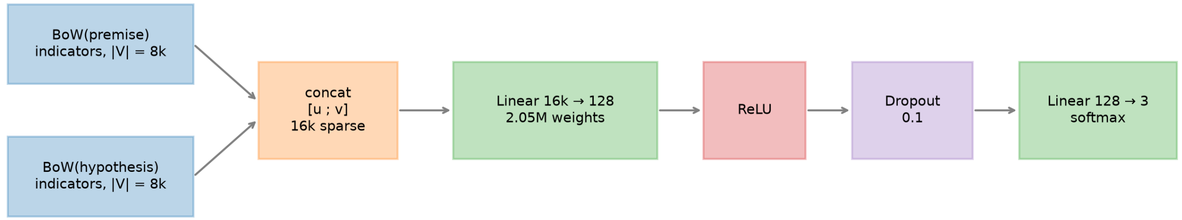


Per-fold CV accuracy, one point per fold (two seeds for the MLPs). Log-linear variants cluster within 0.4 pt; joint pairing and bigrams fall 5 pt below; the MLPs sit 8 pt above.

Q4 · Architectures

- Log-linear (lbfgs, L2), C ∈ {0.1, 1, 10}: flat ≈ 64%
- MLP: 16k → 128 → ReLU → dropout 0.1 → 3
- Adam 1e-3, weight decay 1e-4, early stopping (10% val, patience 3)
- MLP 72.5% (+8 pts, 2 seeds): hidden layer mixes P and H
- Third: residual MLP (LayerNorm + skip), 2 blocks × 128: 72.3 vs 72.5 → P(rope) = 0.99, depth does not help
- Hidden 512: 72.7 vs 72.5 → P(rope) = 1.00, width does not help either; keep 128 (4× fewer params, 5× faster)

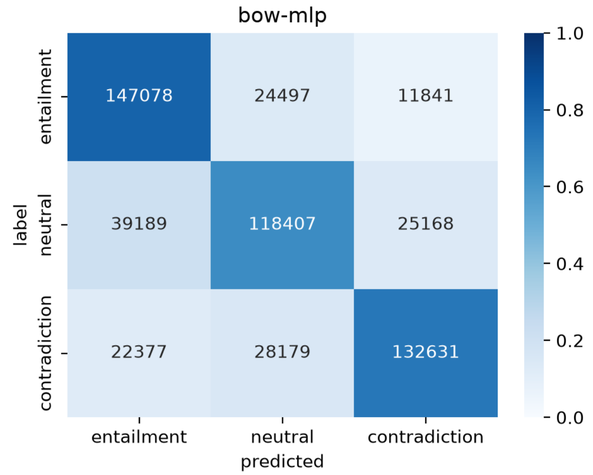
2.05M parameters, 99.98% in the first layer; Adam 1e-3, weight decay 1e-4, batch 64, early stopping on a 10% validation slice (patience 3)



remove the ReLU and the two Linear layers collapse into one 16k → 3 map: exactly the log-linear model (64%).
The hidden layer is where premise and hypothesis features interact (+8 pt).

residual variant: after the ReLU, 2 × [LayerNorm → Linear 128 → ReLU → Dropout] with a skip connection; 72.3 vs 72.5, inside the ROPE

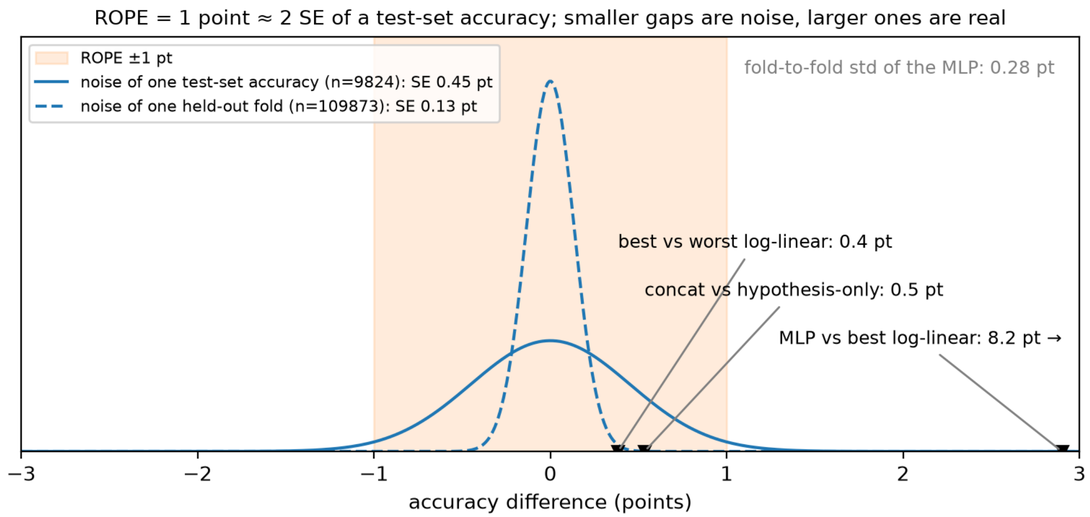
The selected network. Without the ReLU the two Linear maps compose into one 16k → 3 map, i.e. the log-linear model; the hidden layer is the only place premise and hypothesis features interact.



Confusion matrix of the MLP over the pooled held-out folds; neutral is confused with both other classes.

Q5 · Evaluation

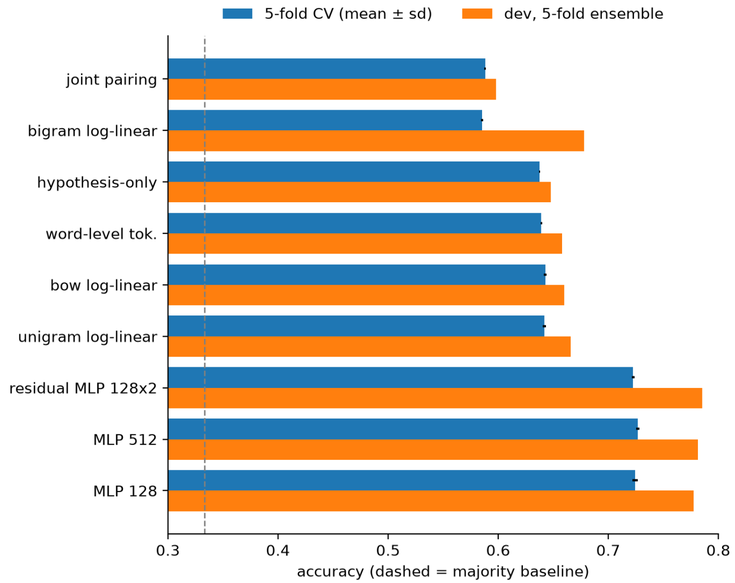
- Stratified 5-fold CV, same folds for all → paired (fold layout on Q1)
- Accuracy (+ macro-F1; balanced, so ≈ equal)
- Baseline: majority 33%, hypothesis-only 63.7%
- Correlated Bayesian t-test (baycomp, ρ = 1/k)
- ROPE = 1 accuracy point ≈ 2 SE of a test-set accuracy (SE ≈ 0.45 pt)
- Neutral is the hardest class



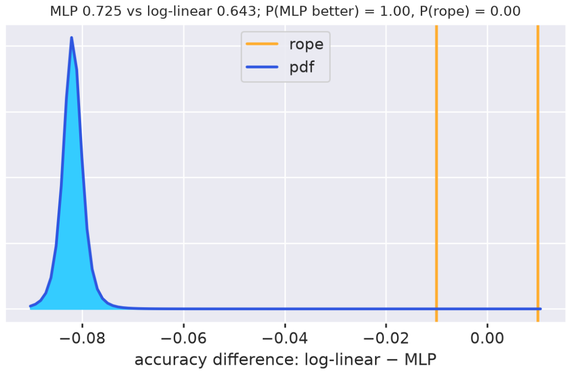
ROPE ±1 pt next to the sampling noise of one evaluation. The log-linear gaps sit inside it; the MLP gap is far outside.

Q6 · Best model

- BoW + MLP (128 hidden): CV 72.5 ± 0.2 over 2 seeds × 5 folds
- P(MLP > log-linear) = 1.00, P(rope) = 0; hidden 512 (72.7) and residual (72.3) inside ROPE → pick the simplest
- Log-linear variants all inside ROPE → pick simplest (C = 0.1, indicators); joint and bigrams clearly worse
- Dev, 5-fold ensemble: MLP 77.8 (512: 78.2, residual 78.5) vs log-linear 66.0
- Test, once: 10 fold models ensembled 77.6 (macro-F1 77.5); a single fold model 74.2



CV mean per experiment next to the dev accuracy of the 5 fold models ensembled; ensembling lifts every model, bigrams most.



Posterior of (log-linear − MLP) accuracy from the correlated t-test: all mass left of the ROPE.